

Disaster Medical Operations — Part 2



CERT Basic Training Unit 4

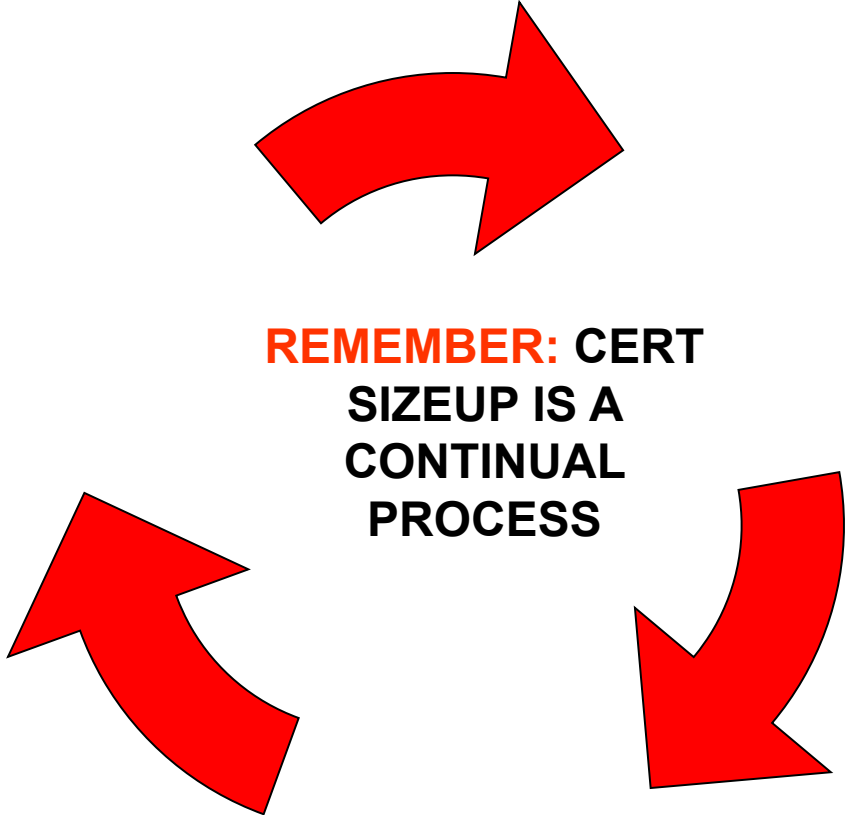
Unit 3 Review

- 3 “Killers”
 - Airway obstruction
 - Excessive bleeding
 - Shock



CERT Sizeup

1. Gather Facts
2. Assess Damage
3. Consider Probabilities
4. Assess Your Situation
5. Establish Priorities
6. Make Decisions
7. Develop Plan of Action
8. Take Action
9. Evaluate Progress



**REMEMBER: CERT
SIZEUP IS A
CONTINUAL
PROCESS**



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Unit Objectives

- Take appropriate sanitation measures to help protect public health
- Perform head-to-toe patient assessments
- Establish a treatment area
- Apply splints to suspected fractures and sprains
- Employ basic treatments for other injuries

Unit Topics

- Public Health Considerations
- Functions of Disaster Medical Operations
- Establishing Medical Treatment Areas
- Conducting Head-to-Toe Assessments
- Treating Fractures, Dislocations, Sprains, and Strains
- Nasal Injuries
- Treating Cold-Related Injuries
- Treating Heat-Related Injuries
- Bites and Stings



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CERT Basic Training

Unit 4: Disaster Medical Operations – Part 2

4-5



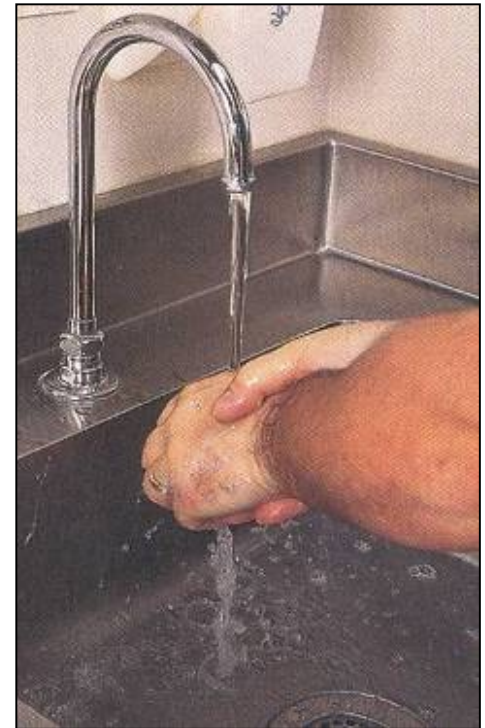
Public Health Considerations

- Maintaining proper hygiene
- Maintaining proper sanitation
- Purifying water (if necessary)
- Preventing spread of disease



Maintaining Hygiene

- Wash hands frequently
 - Or use alcohol-based hand sanitizer
- Wear non-latex exam gloves
- Wear N95 mask and goggles
- Keep dressings sterile
- Avoid contact with body fluids
 - “If it is warm, wet, and not yours, don’t touch it!”



Maintain Sanitation

- Control disposal of bacterial sources
- Put waste products in plastic bags
 - Tie off bags and mark them as medical waste
- Bury human waste



Water Sanitation Methods

- Boil water for 1 minute
- Water purification tablets
- Non-perfumed liquid bleach
 - 8 drops/gal of water
 - 16 drops/gal if water is cloudy
 - Let stand for 30 minutes before use



Functions of Disaster Medical Operations

- Triage
- Treatment
- Transport
- Morgue
- Supply



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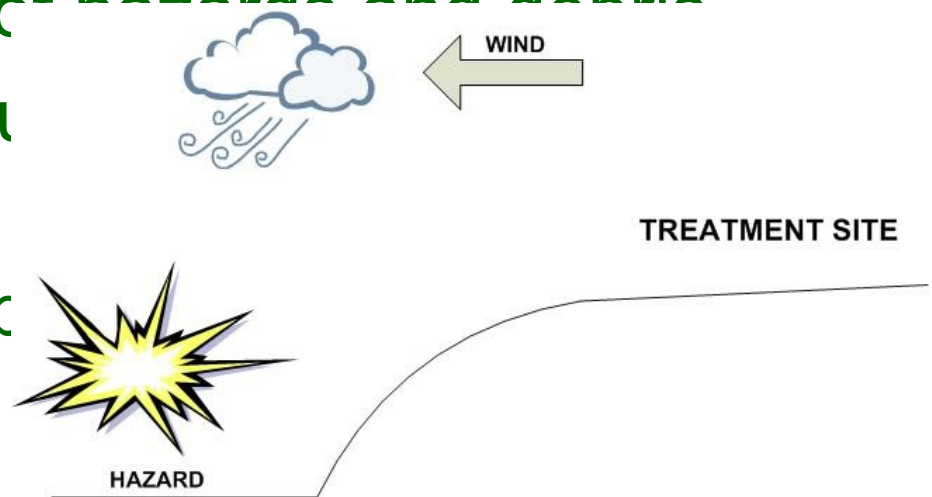
Establish a Medical Treatment Area

- Select site and set up treatment area as soon as injured survivors are confirmed
- When determining best location(s) for treatment area, consider:
 - Safety of rescuers and survivors
 - Most effective use of resources



Treatment Area Site Selection

- The site selected should be:
 - In a safe area, free of hazards and debris
 - Upwind, uphill, and up from hazard zone(s)
 - Accessible by transport
 - Expandable



The treatment site should be uphill and upwind from the hazard.

Most Effective Use of CERT Resources

- To help meet the challenge of limited resources, CERT may need to establish:
 - Decentralized medical treatment location (more than one location)
 - Centralized medical treatment location (one location)

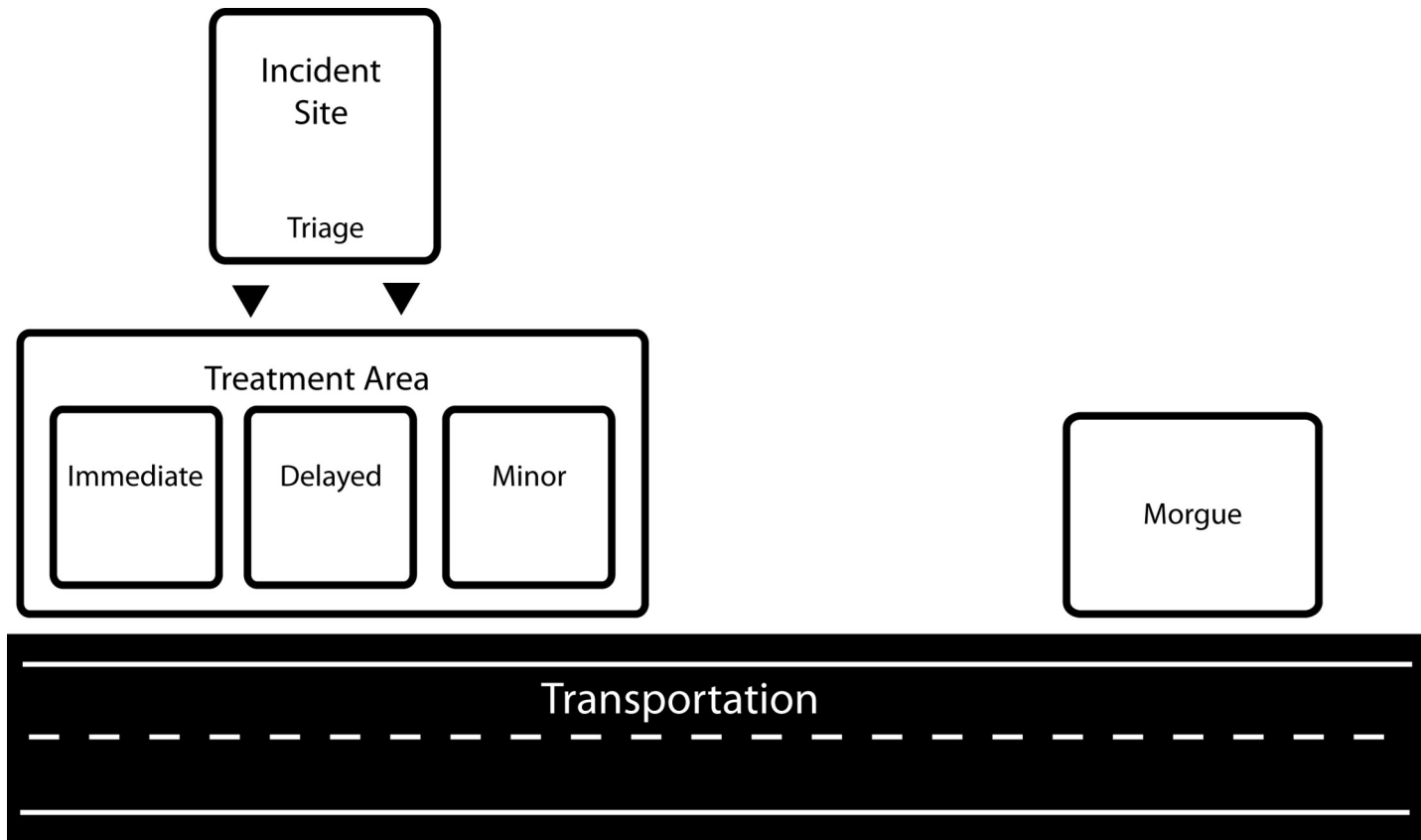


Treatment Area Layout

- Four treatment areas:
 - “I” for Immediate care
 - “D” for Delayed care
 - “M” for Minor injuries/walking wounded
 - “DEAD” for the morgue



Treatment Area Layout



Treatment area layout, showing the organization for the incident site, triage, transportation, and morgue



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Treatment Area Organization

- Assign treatment leader to each treatment area
- Document thoroughly
 - Available identifying information
 - Description (age, sex, body build, estimated height)
 - Clothing
 - Injuries
 - Treatment
 - Transfer location



Head-to-Toe Assessment

- Objectives of head-to-toe
 - Determine extent of injury
 - Determine type of treatment needed
 - Document injuries



DCAP-BTLS

- Deformities
- Contusions
- Abrasions
- Punctures
- Burns
- Tenderness
- Lacerations
- Swelling



▲ DEFORMITIES



▲ CONTUSIONS



▲ ABRASIONS



▲ PUNCTURES/PENETRATIONS



▲ BURNS



▲ TENDERNESS



▲ LACERATIONS



▲ SWELLING

Where and When

- Light damage: assess in place
- Moderate damage: move to treatment area first
- Assess and tag everyone
- Both verbal and hands on

Conducting Head-to-Toe Assessment

- Pay careful attention
- Look, listen, and feel
- Check own hands for patient bleeding
- If you suspect a spinal injury in unconscious survivors, treat accordingly
- Check PMS in all extremities
- Look for medical identification

Order of Assessment

1. Head
2. Neck
3. Shoulders
4. Chest
5. Arms
6. Abdomen
7. Pelvis
8. Legs



Closed-Head, Neck, Spinal Injuries

- Do no harm
 - Minimize movement of head and neck
- Keep spine in straight line
- Stabilize head

Treating Burns

- Conduct thorough sizeup
- Treat with first aid
 - Cool burned area
 - Cover with sterile cloth to reduce risk of infection



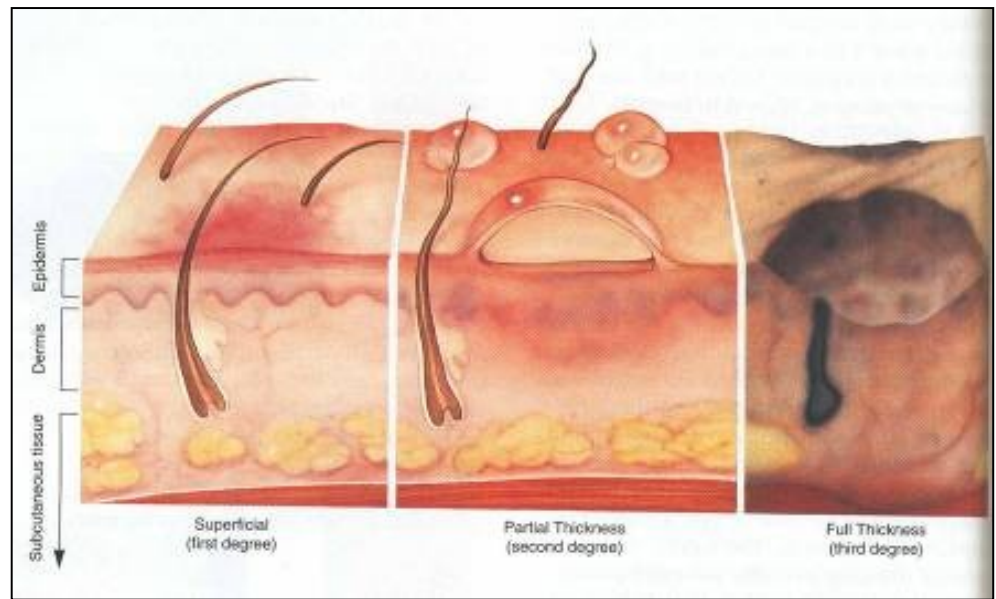
Burn Severity

- Factors that affect burn severity:
 - Temperature of burning agent
 - Period of time survivor exposed
 - Area of body affected
 - Size of area burned
 - Depth of burn



Burn Classifications

- Superficial: epidermis
- Partial Thickness: dermis and epidermis
- Full Thickness: subcutaneous layer and all layers above



Burn Treatment: DOs

- When treating a burn survivor, DO:
 - Cool skin or clothing if they are still hot
 - Cover burn loosely with dry, sterile dressings to keep air out, reduce pain, and prevent infection
 - Elevate burned extremities



Burn Treatment: DON'Ts

- When treating a burn survivor, DO NOT:
 - Use ice
 - Apply antiseptics, ointments, or other remedies
 - Remove shreds of tissue, break blisters, or remove adhered particles of clothing



Treatment for Chemical Burns

- Remove cause of burn + affected clothing/jewelry
- If irritant is dry, gently brush away as much as possible
 - Always brush away from eyes, survivor, and you
- Flush with lots of cool running water
- Apply cool, wet compress to relieve pain
- Cover wound loosely with dry, sterile or clean dressing
- Treat for shock if appropriate



Inhalation Burns Signs and Symptoms

- Sudden loss of consciousness
- Evidence of respiratory distress or airway obstruction
- Soot around mouth or nose
- Singed facial hair
- Burns around face or neck



Figure 27-26 A singed mustache and burns to the tip of the tongue signal danger of airway burns or burns to the eyes.

Wound Care

- Control bleeding
- Clean wound
- Apply dressing and bandage



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Cleaning and Bandaging Wounds

- Clean by irrigating with clean, room temperature water
 - NEVER use hydrogen peroxide
 - Irrigate but do not scrub
- Apply dressing and bandage
 - Dressing applied directly to wound
 - Bandage holds dressing in place

Rules of Dressing

- If active bleeding:
 - Redress OVER existing dressing
- If no active bleeding:
 - Remove bandage and dressing to flush wound
 - Check for infection every 4-6 hours

Signs of Infection

- Signs of possible infection
 - Swelling around wound
 - Discoloration
 - Discharge from wound
 - Red striations from wound site



Amputations

- Control bleeding; treat shock
- If amputated body part is found:
 - Save tissue parts, wrapped in clean material and placed in plastic bag
 - Keep tissue parts cool, but NOT directly on ice
 - Keep severed part with survivor



Impaled Objects

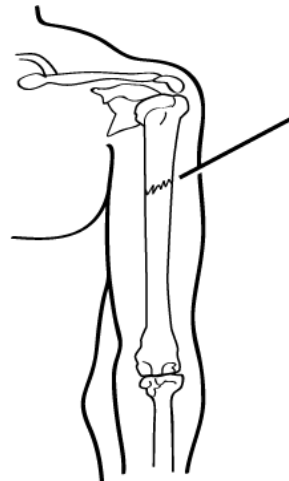
- When foreign object is impaled in patient's body:
 - Immobilize affected body part
 - Do not attempt to move or remove
 - Try to control bleeding at entrance wound
 - Clean and dress wound, making sure to stabilize impaled object



Fractures, Dislocations, Sprains, Strains

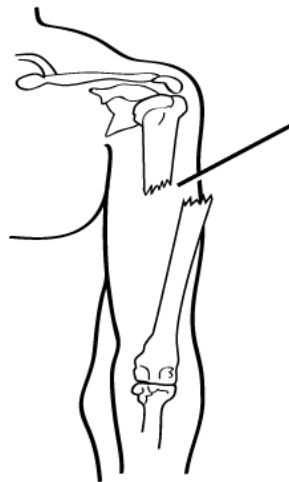
- Immobilize injury and joints immediately above and below injury site
- If uncertain of injury type, treat as fracture

Types of Fractures



Closed Fracture

Closed Fracture in which the fracture does not puncture the skin.



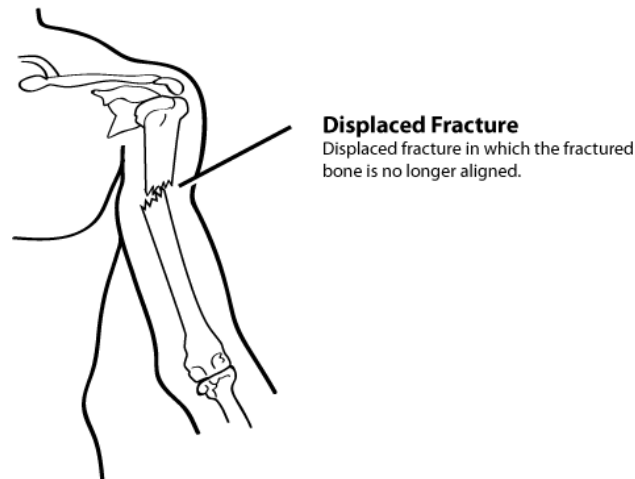
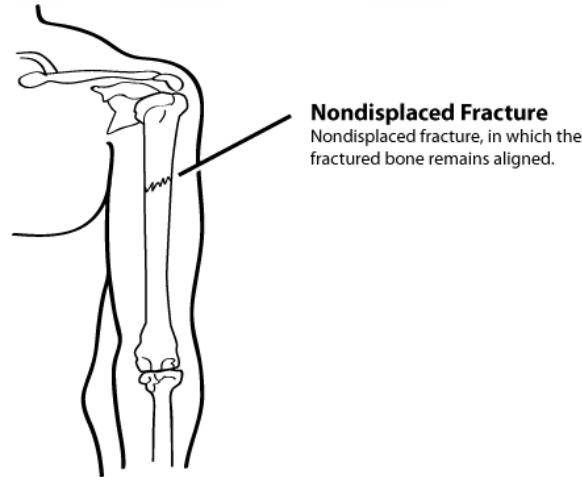
Open Fracture

Open Fracture in which the bone protrudes through the skin.

Treating Open Fractures

- Do not draw exposed bone ends back into tissue
- Do not irrigate wound
- Cover wound with sterile dressing
- Splint fracture without disturbing wound
- Place moist dressing over bone end

Displaced and Nondisplaced Fractures



Dislocations

- Dislocation is injury to ligaments around joint
 - So severe that it permits separation of bone from its normal position in joint
- Treatment
 - Immobilize; do NOT relocate
 - Check PMS before and after splinting/immobilization

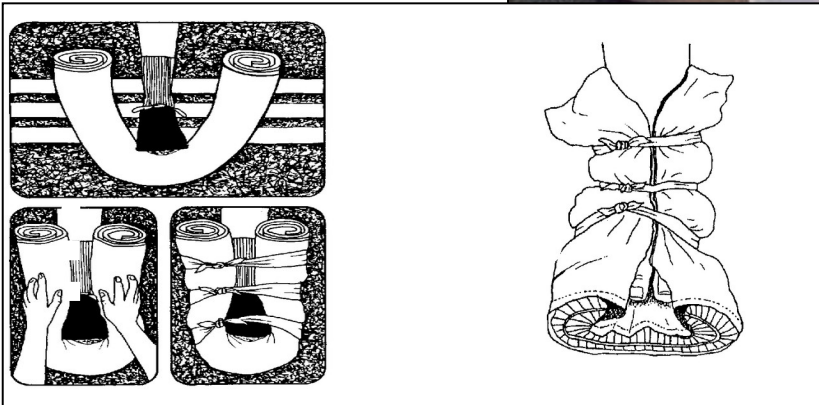
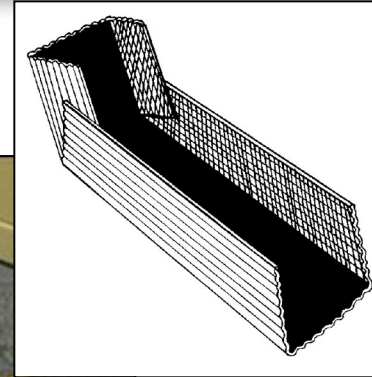
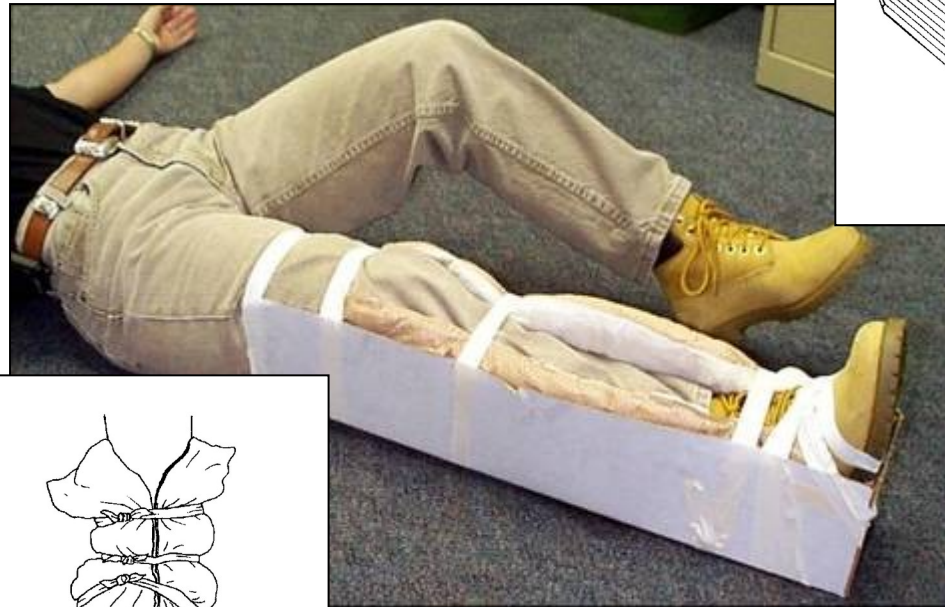
Signs of Sprain

- Tenderness at site
- Swelling and bruising
- Restricted use or loss of use



Damaged vessels from an ankle sprain can cause bruising

Splinting



Splinting Guidelines

1. Support injured area above and below injury
2. Assess PMS in extremity
3. Splint injury in position that you find it
4. Don't try to realign bones or joints
5. Fill voids to stabilize and immobilize
6. Immobilize above and below injury
7. After splinting, reassess PMS

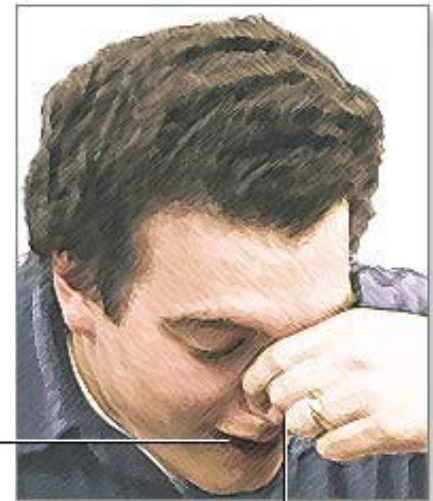
Nasal Injuries

- Causes
 - Blunt force to nose
 - Skull fracture
 - Nontrauma conditions, e.g., sinus infections, high blood pressure, and bleeding disorders
- Cautions
 - Large blood loss from nosebleed can lead to shock
 - Actual blood loss may not be evident because survivor will swallow some amount of blood

Treatment of Nasal Injuries

- Control nasal bleeding:
 - Pinch nostrils or put pressure on inner line under nose
 - Have survivor sit with back
- Ensure that airway re
- Keep survivor calm

Sit and lean forward slightly



Breathe through mouth

Pinch nostrils

ADAM.

Cold-Related Injuries

- Hypothermia :
 - Occurs when body's temperature drops below normal
- Frostbite:
 - Occurs when extreme cold shuts down blood flow to extremities, causing tissue death



Symptoms of Hypothermia

- Body temperature of 95° F or lower
- Redness or blueness of skin
- Numbness and shivering
- Slurred speech
- Unpredictable behavior
- Listlessness



Hypothermia Treatment

- Remove wet clothing
- Wrap survivor in blanket
- Protect survivor from weather
- Provide food and drink to conscious survivors
- Do not attempt to massage to warm body
- Place unconscious survivor in recovery position
- Place survivor in warm bath

Symptoms of Frostbite

- Skin discoloration
- Burning or tingling sensation
- Partial or complete numbness



FIGURE 29-8 ▲ Edema and blister formation 24 hours after frostbite injury in an area covered by a tightly fitted boot. (From Auerbach PS: *Wilderness medicine*, ed 4, St Louis, 2001, Mosby.)

Frostbite Treatment

- Immerse injured area in warm (NOT hot) water
 - Warm slowly!
- Do NOT allow part to re-freeze
- Do NOT attempt to use massage
- Wrap affected body parts in dry, sterile dressing



Heat-Related Injuries

- Heat cramps:
 - Muscle spasms brought on by over-exertion in extreme heat
- Heat exhaustion:
 - Occurs when exercising or working in extreme heat results in loss of body fluids
- Heat stroke:
 - Survivor's temperature control system shuts down
 - Body temperature rises so high that brain damage and death may result

Symptoms of Heat Exhaustion

- Cool, moist, pale or flushed skin
- Heavy sweating
- Headache
- Nausea or vomiting
- Dizziness
- Exhaustion



Symptoms of Heat Stroke

- Hot, red skin
- Lack of perspiration
- Changes in consciousness
- Rapid, weak pulse and rapid, shallow breathing

Treatment of Heat-Related Injuries

- Remove from heat to cool environment
- Cool body slowly
- Have the survivor drink water, SLOWLY
- No food or drink if survivor is experiencing vomiting, cramping, or is losing consciousness

Treatment for Bites/Stings

- If bite or sting is suspected, and situation is non-emergency:
 - Remove stinger if still present by scraping edge of credit card or other stiff, straight-edged object across stinger
 - Wash site thoroughly with soap and water
 - Place ice on site for 10 minutes on and 10 minutes off



Anaphylaxis

- Check airway and breathing
- Calm individual
- Remove constrictive clothing and jewelry
- Find and help administer survivor's Epi-pen
- Watch for signs of shock and treat appropriately



Unit Summary

- Public health concerns related to sanitation, hygiene, and water purification
- Organization of disaster medical operations
- Establishing treatment areas
- Conducting head-to-toe assessments
- Treating wounds, fractures, sprains, and other common injuries



Homework Assignment

- Read unit to be covered in next session
- Bring necessary supplies for next session
- Wear appropriate clothes for next session
- Practice complete head-to-toe assessment on friend or family member

